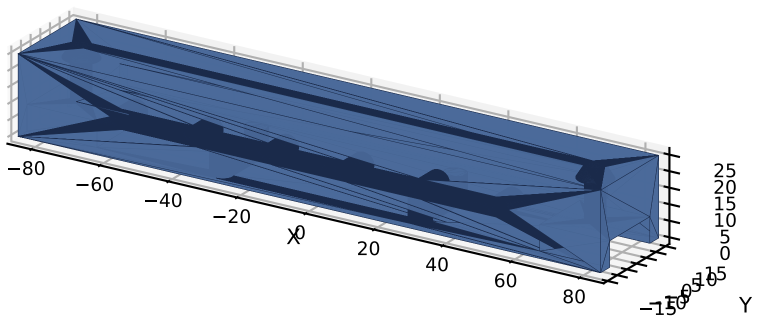
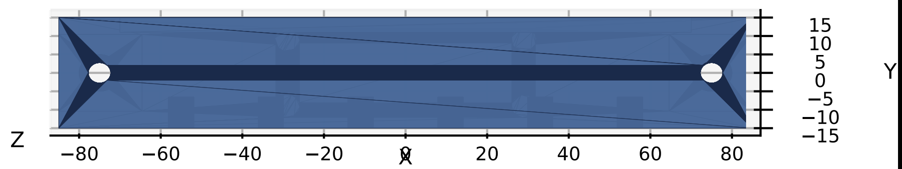
CSM V3 -- HMI Module 10 -- Cross Beam V1.0

*Horizontal cross-member joining mast tops * Carries Pi 4 + future accessories * PETG, 100% infill, 4 walls*

Isometric View



Top View (X-Y) -- mast pockets + Pi inserts

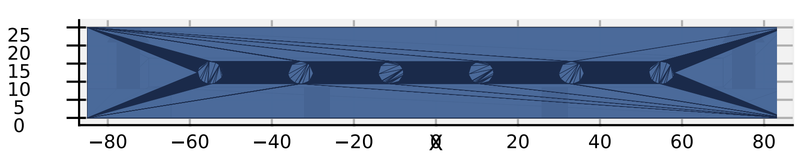


PART	Cross Beam
VERSION	V1.0
MODULE	10 -- HMI
QTY	1
MATERIAL	PETG (PA12 prod)
INFILL	100%, 4 walls
LAYER	0.2 mm
MASS	~143 g
TIME	~6-8 h
INSTALL	top of HMI masts (X=0)

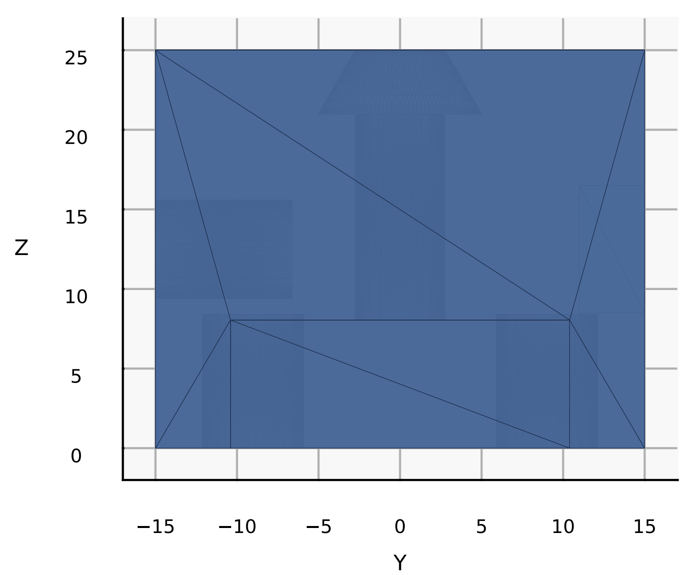
DIMENSIONS (mm)

Beam L x W x H	170.0 x 30.0 x 25.0
Mast spacing	150.0 (X=+/-75)
Mast pockets	20.8 sq x 8.0 deep
Mast bolt	M5 x 5.5 clear, csk 10.0
Pi mount PCD	58.0 x 18.0 (4x M5)
Cable channel	8.0 x 4.0 (rear)
Insert pocket	6.2 dia x 8.5 deep
Front inserts	6 @ 22.0 pitch

Front View (-Y face) -- 6x accessory inserts



Side View (+X face) -- end profile + cable channel



ASSEMBLY

1. Print PETG horizontal on bed, 100% infill, 4 walls, 0.2 mm layers
2. Heat 4x M5 brass inserts into BOTTOM face pockets (Pi mount points), 220 C ~30 s each
3. Heat 6x M5 brass inserts into FRONT face pockets (future accessories)
4. Position beam over both mast tops; bottom-face pockets engage mast XS for anti-rotation
5. M5 socket-head bolts (~25 mm) through top countersinks, self-tap into 2020 center bore
6. Verify beam is level; check Pi mount accessible
7. Mount Pi Carrier V1.0 (Part 3) to BOTTOM inserts
8. Route cables up REAR mast face into cable channel

EXPANSION (6x M5 inserts, FRONT face):
- Status LED panel - Microphone - Motion sensor
- OLED text display - Camera mount - Future UI

CONSTRAINTS:
- Pi PCD Y = 18 mm (limited by beam width 30)
- Standard Pi 4 PCD is 49 mm - Pi Carrier V1.0 will adapt 18 mm beam pitch to 49 mm Pi pitch